Esercizio.

Check that the following are expressible as Δ_0-formulae.

Dimostrazione.

- $x \subseteq y$: $\forall z \in x(z \in y)$
- $x = \{y, z\}$: $\forall t \in x(t = y \vee t = z) \wedge \exists t \in x(t = y) \wedge \exists tinx(t = z)$
- $x = (y, z)$: $\forall t \in x(t = \{y\} \vee t = \{y, z\}) \wedge \exists t \in x(t = \{y\}) \wedge \exists t \in x(t = \{y, z\})$
- $x = \bigcup y$: $\forall t \in x \exists z \in y(t \in z) \wedge \forall z \in y \forall t \in z(t \in x)$
- $x = \bigcap y$: $\forall t \in x \exists r \in y(t \in r) \wedge \forall t \in y \forall z \in t(z \in x)$
- $x = y \setminus z$: $\forall t \in x(t \in x \wedge t \notin z) \wedge \forall t \in y(t \notin z \to t \in x)$
- $x = y \times z$: $\forall t \in x \exists a \in y \exists b \in z(t = (a, b)) \wedge \forall a \in y \forall b \in z \exists r \in x(r = (a, b))$
- x is binary relation: $\forall t \in x \exists p \in x \exists y \in p \exists q \in x \exists z \in q(t \in y \times z)$
- x function: x binary relation and other stuff ..
- x surjection: x function and other stuff..
- x bijection: x surjection and x injection
- $x = dom(y)$: y is function $\wedge \forall t \in x \exists u \in y \exists z \in u((t, z) \in y) \wedge \forall u \in y \forall z \in u \forall a \in y \forall b \in a\Big((b, z) \in y \to \exists t \in x((b, z) = t)\Big)$
- $x = ran(y)$: analogue
- x n-ary relation: analogue
- x trans: $\forall y \in x \forall z \in y(z \in x)$
- $x = \emptyset$: $\forall y \in x(y \neq y)$
- $x \in Ord$: $\forall y \in x \forall z \in y(trans(y) \wedge trans(z))$
- x successor ordinal: $x \in Ord \wedge \exists y \in x \forall z \in x(z \in y \vee z = y)$
- x limit ordinal: $x \in Ord \wedge$ not successor
- Let $n \in \omega$. Let S denote the successor function (whose graph is clearly $Delta_0$. Then $x = n$: $x = S(..n\ times..S(\emptyset)..)$.
- $x = \omega$: $x \in Ord \wedge x$ limit $\wedge \forall y \in x(y$ successor $\vee\, y = \emptyset)$

$\square$